

BARANOVICHI, Belarus

WMO: 269410

Lat: 53.1319N

Lon: 25.9719E

Elev: 193

StdP: 99.02

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.5	-16.2	-22.3	0.5	-18.3	-19.0	0.7	-15.7	9.4	-0.1	8.4	-0.8	2.0	30	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	28.8	19.6	27.1	19.0	25.5	18.1	21.0	26.5	20.1	25.1	19.2	23.9	3.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.2	23.5	18.2	13.5	22.6	17.4	12.7	21.6	61.7	26.6	58.4	25.1	55.3	23.9	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.1	6.2	-24.1	30.9	3.1	1.4	-26.3	31.9	-28.1	32.7	-29.8	33.5	-32.0	34.6		
(5)			-24.3	22.9	3.0	1.2	-26.5	23.8	-28.3	24.6	-30.0	25.3	-32.2	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-4.2	-3.1	1.3	8.3	13.4	16.5	19.0	17.9	12.8	7.2	2.2	-2.3		
	DBStd	9.28	5.95	5.76	4.65	4.37	4.22	3.41	3.14	3.08	3.41	4.08	4.57	5.47		
	HDD10.0	1914	441	368	272	83	16	0	0	0	13	106	236	381		
	HDD18.3	4083	699	601	529	302	162	76	31	45	168	346	485	639		
	CDD10.0	987	0	0	1	31	120	195	278	246	97	19	1	0		
	CDD18.3	113	0	0	0	1	8	21	50	32	2	0	0	0		
	CDH23.3	902	0	0	0	6	74	159	402	243	17	0	0	0		
	CDH26.7	186	0	0	0	1	11	23	97	54	1	0	0	0		
(14) Wind	WSAvg	3.1	3.4	3.4	3.4	3.1	3.0	2.8	2.6	2.5	2.7	3.1	3.3	3.3		
(15) Precipitation	PrecAvg	610	34	28	35	40	56	80	81	68	53	48	46	41		
	PrecMax	786	86	66	76	103	115	145	204	210	176	142	92	82		
	PrecMin	440	9	6	4	12	20	31	14	8	10	2	13	13		
	PrecStd	96	18	15	16	21	25	31	51	41	34	33	20	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	8.4	15.1	24.0	28.2	29.2	31.4	30.9	25.5	20.3	13.3	8.3		
		MCWB	5.8	5.8	9.1	14.9	18.1	19.5	20.7	21.0	16.7	15.5	10.6	7.1		
	2%	DB	5.0	6.2	12.2	20.6	25.4	27.2	29.4	28.0	22.8	17.0	11.0	6.5		
		MCWB	3.8	4.5	7.2	12.2	17.1	19.0	20.0	19.7	15.9	13.2	9.6	5.6		
	5%	DB	3.7	4.7	9.8	18.1	23.2	25.3	27.4	26.0	20.6	14.7	9.3	5.2		
		MCWB	2.8	3.4	6.0	11.1	15.7	17.7	19.5	18.3	14.9	12.1	8.1	4.4		
	10%	DB	2.3	3.3	7.6	15.8	21.1	23.4	25.6	24.1	18.8	13.0	8.0	3.7		
		MCWB	1.6	2.3	4.8	9.8	14.3	17.1	18.7	17.5	14.1	11.1	7.1	3.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	6.5	9.8	15.4	20.2	21.2	22.9	21.9	18.1	15.7	11.2	7.4		
		MCDB	6.8	8.0	13.9	23.2	26.6	26.7	28.2	28.5	23.2	18.9	12.6	8.0		
	2%	WB	4.0	4.9	7.9	13.1	18.1	19.9	21.4	20.6	16.7	13.9	9.6	5.8		
		MCDB	4.7	5.9	11.2	18.5	23.4	25.2	27.0	26.2	21.1	16.3	10.8	6.4		
	5%	WB	2.9	3.5	6.4	11.7	16.6	18.9	20.4	19.7	15.7	12.5	8.3	4.5		
		MCDB	3.6	4.4	9.1	16.5	21.8	23.8	25.3	24.4	19.3	14.1	9.2	5.1		
	10%	WB	1.7	2.3	5.1	10.5	15.2	17.7	19.4	18.5	14.8	11.2	7.1	3.0		
		MCDB	2.3	3.2	7.4	14.8	19.9	22.0	24.1	22.3	18.0	13.0	7.8	3.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.8	6.4	8.7	9.7	9.0	9.5	9.5	8.6	6.3	3.9	3.7		
		MCDBR	3.7	5.1	9.9	12.4	12.5	11.7	12.1	12.2	11.9	8.9	5.6	4.2		
		MCWBR	3.5	4.0	6.0	5.9	5.8	5.2	4.8	4.9	6.0	5.7	4.4	3.8		
		MCDBR	3.7	4.9	8.7	10.5	11.1	10.0	10.4	9.7	9.8	7.8	5.0	4.1		
	5% WB	MCWBR	3.6	4.1	5.8	5.6	5.7	5.1	4.8	4.5	5.5	5.4	4.2	3.7		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.330	0.364	0.413	0.403	0.391	0.424	0.420	0.383	0.354	0.337	0.297		
	taud		2.186	2.140	2.186	2.178	2.275	2.335	2.255	2.264	2.328	2.387	2.286	2.227		
	Ebn at Noon		663	749	808	814	846	860	821	801	786	730	610	586		
	Edh at Noon		67	96	114	131	125	119	127	119	99	76	62	54		
(44) All-Sky Solar Radiation	RadAvg		0.71	1.48	2.77	4.12	5.26	5.62	5.28	4.58	3.12	1.64	0.62	0.44		
	RadStd		0.12	0.27	0.42	0.49	0.51	0.42	0.51	0.39	0.36	0.33	0.07	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	+1.18	+1.47	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.44	N/A	N/A	N/A	N/A	+0.77	N/A	N/A	+90	N/A

Nomenclature: See separate page